



Loan Default Risk Prediction

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1. Problem and Data

Home Credit lends to people with little or no credit history. Our goal was to predict, at the time of application, how likely each applicant is to default. The Kaggle data has seven linked tables: the current application plus six tables of history, covering previous Home Credit loans, loans at other banks, credit card balances, point of sale balances and installment payments. Only 8.1% of applicants default, so the data is heavily imbalanced.

2. Data Cleaning and Joining

We combined the train (307,511 rows) and test (48,744 rows) application files so both were cleaned the same way. Many columns were badly incomplete, some over 60% empty (Figure 1). We grouped columns by missingness (under 20%, 20 to 50%, over 50%) and chose a fix for each type. Numeric columns got the median. Columns where missing clearly meant "none", like car age for people without a car, got 0. Categorical columns got a "Missing" label. All fill values came from the training rows only, to avoid data leakage.

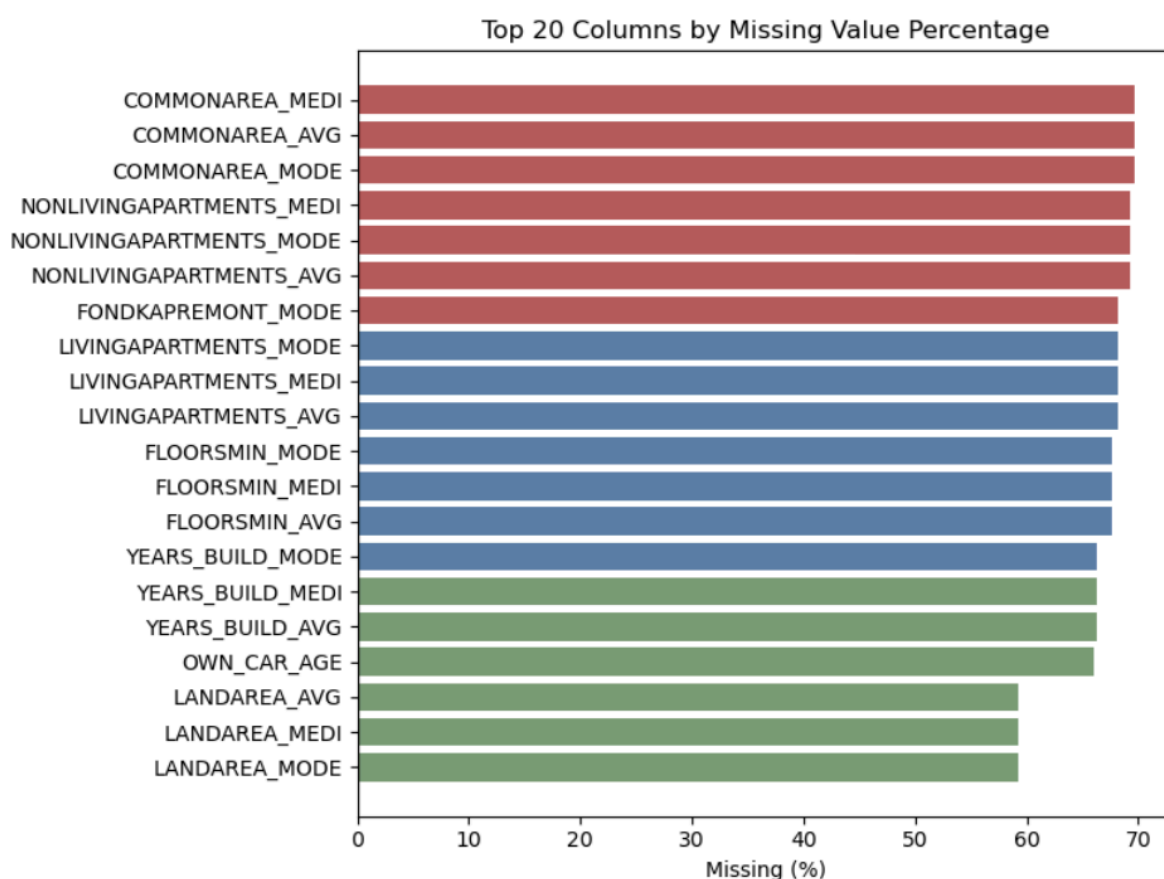


Figure 1: Columns with the most missing values

The history tables have many rows per person, so each was summarised to one row per applicant with counts, means, sums and maximums, then added with a left join. bureau_balance

has no applicant ID, so it was summarised per loan and joined through bureau first. Two interest rate columns over 99% empty were dropped. DAYS columns missing only for refused or canceled loans were set to 0, since that stage never happened. The placeholder 365243 (about 1,000 years) was replaced with a missing value. The final table had 356,255 rows and 168 columns.

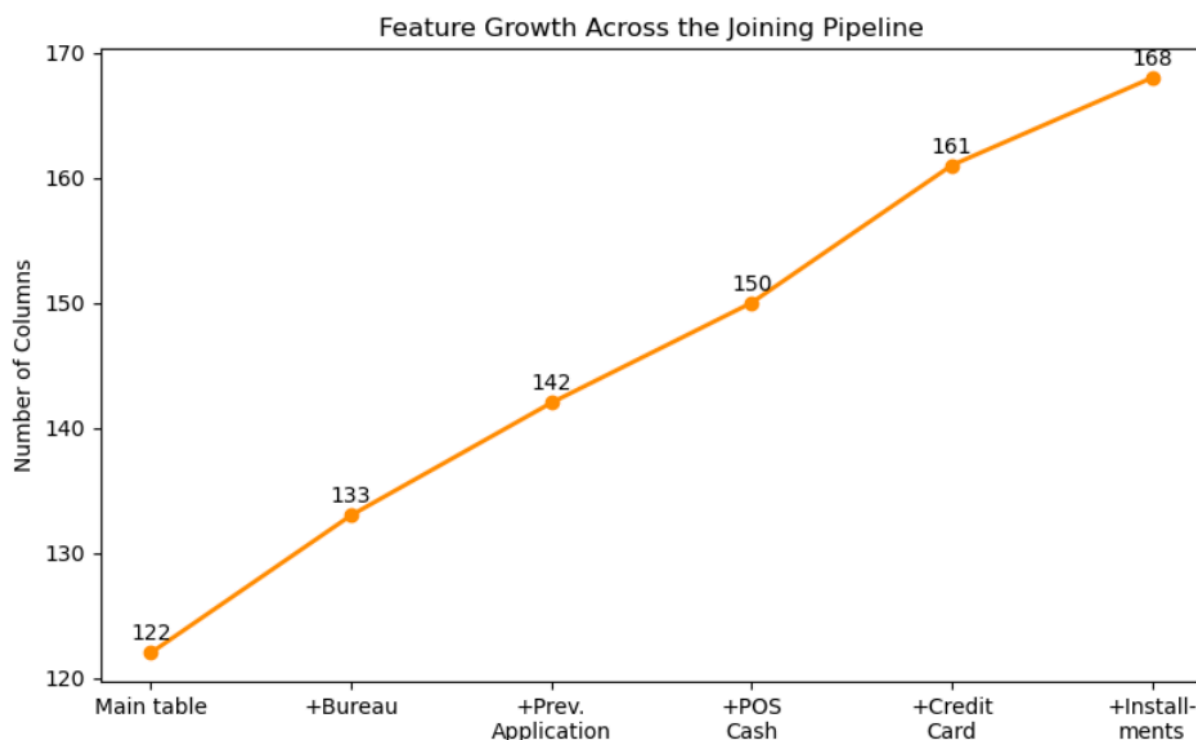


Figure 2: Columns added as each table was joined

3. Feature Engineering

Raw totals say little on their own. A 500,000 loan is easy for a high earner and a burden for a low one. So we built 29 new features that describe how people handle money. Affordability features include credit to income, annuity to income, credit to annuity (roughly the loan length) and credit to goods price. We combined the three external credit scores into a mean, minimum, maximum and spread. From the history tables we built the bureau debt ratio, credit received versus amount requested, credit card utilisation (balance divided by limit), and the installment payment ratio and delay. We also added flags for whether an applicant has any history in each table, because no history is not the same as a clean one. DAYS_EMPLOYED also had the 365243 placeholder for 18% of applicants, so we flagged it and set it to be missing.

Statistics backed this up. The mean external score showed the biggest gap between defaulters and repayers (Cohen's d of -0.81). Credit card utilisation (0.47) and bureau debt ratio (0.33) came next. Because the features are skewed, we used the Mann-Whitney U test. Nearly everything was significant with 300,000 rows, so we judged importance by effect size instead. Chi-square tests showed education, occupation, income type and gender were linked to default.

4. Modelling and Validation

We split the labelled data into 80% training and 20% test, stratified to keep the 8% default rate. The test set was only used once, at the end. Missing values were filled with training medians, so nothing leaked from the test set. Model settings were chosen on a validation set taken from the training data, never on the test set.

We used ROC-AUC, not accuracy. Predicting "no default" for everyone would give 92% accuracy and be useless. ROC-AUC measures how well the model ranks defaulters above repayers, where 0.5 is guessing and 1.0 is perfect. We started with a logistic regression baseline on the application table only. We then trained a random forest and a gradient boosting model (HistGradientBoostingClassifier) on all seven tables, and tested three learning rates for gradient boosting.

5. Results

All scores below are on the test set, which the models had never seen.

Model	Features used	Test ROC-AUC
Logistic Regression (baseline)	Application table only	0.749
Random Forest	All 7 tables	0.763
Gradient Boosting (final)	All 7 tables	0.782

Adding the engineered features and the six history tables raised ROC-AUC by **4.4%** over the baseline. Our final score on unseen data is **0.782**, above the 0.75 target.

Improvement over baseline: 0.0339

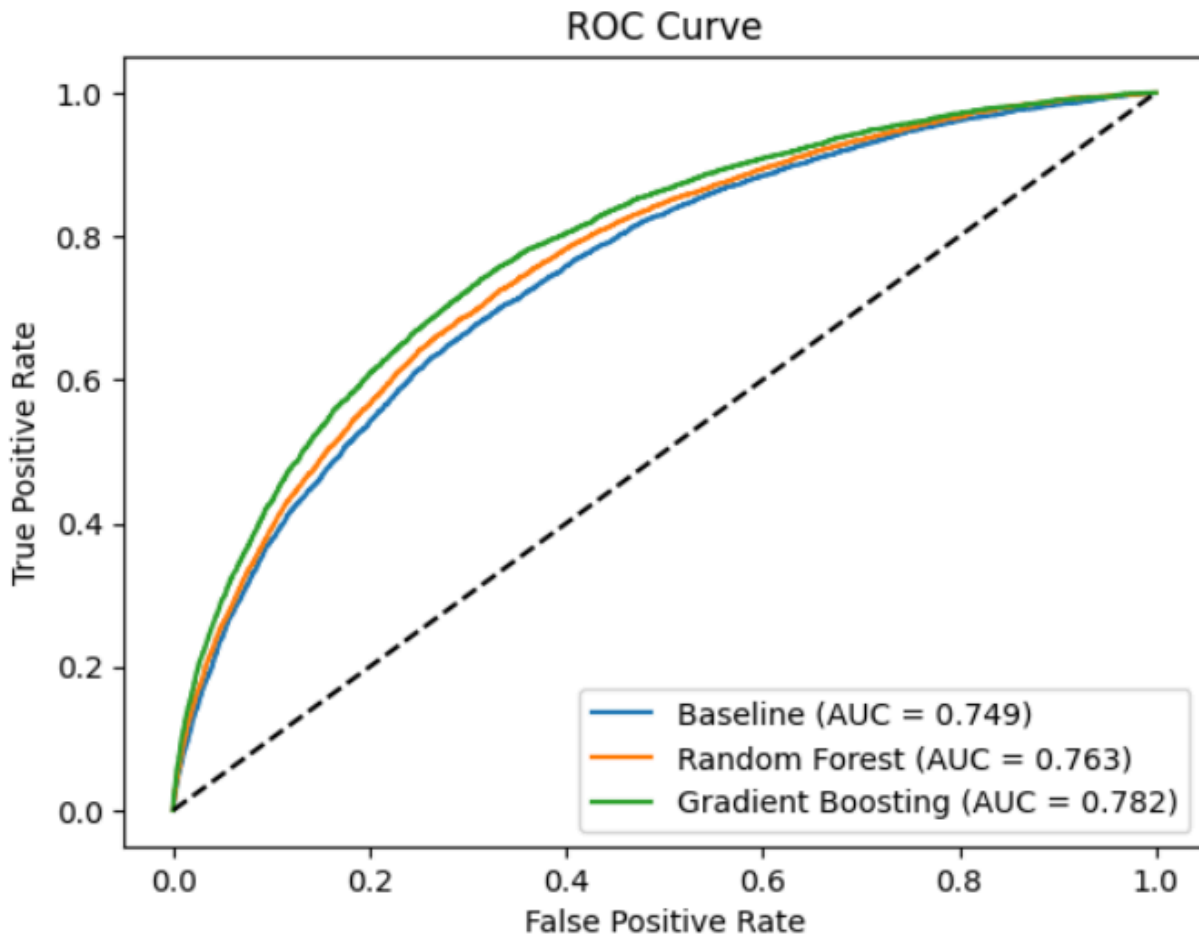


Figure 3. ROC AUC Curve

6. Most Predictive Features

We measured importance by shuffling each feature and seeing how much ROC-AUC dropped. The mean external credit score was by far the strongest, about fourteen times more important than anything else. Next came loan structure (annuity, credit to goods ratio, credit to annuity ratio) and past repayment behaviour (payment ratio, total paid, installments left). Bureau debt ratio, credit card utilisation and age also ranked highly. Seven of the top fifteen features were ones we engineered, and most came from the history tables. This makes sense for lending. People with low scores, heavy loan burdens, weak repayment records or maxed out cards are the ones most likely to default.

The external score dominates because it already sums up a person's credit record from outside agencies. Still, the model does not rely on it alone. The loan structure features show that how a loan is set up matters, not just its size. A borrower who takes a large loan with high yearly payments is more stretched than one repaying slowly over many years. Repayment history works like a track record. Someone who underpaid or paid late before is likely to struggle again. Utilisation and bureau debt show how much pressure a person is already under before this new loan. Age likely reflects stability, since younger applicants tend to have lower incomes and

shorter job histories. For a lender, this means the model can do more than approve or reject. It can point to the reason an applicant looks risky, such as a heavy repayment burden or a history of late payments. That could help staff review borderline cases or offer a smaller loan instead of a flat rejection.

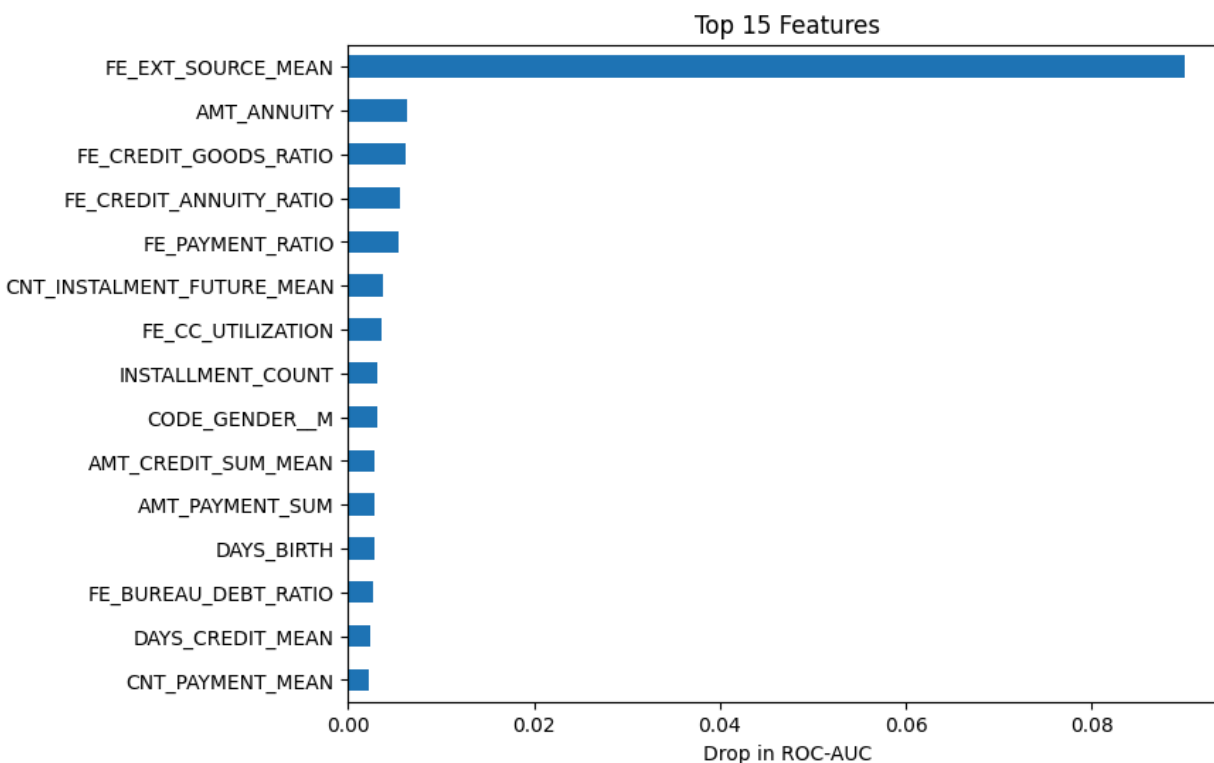


Figure 4. Top 15 Features Graph

7. Limitations and Next Steps

Our model has a few limitations. First, it finds patterns in the data, but it cannot explain why a person defaults. A high credit card balance may go along with default without actually causing it. Second, some predictive features raise fairness concerns. Gender and age both helped the model, but using them to decide who gets a loan may be unfair, and in many countries it is illegal. A real lender would need to review these features, or remove them and check how much performance drops. Third, applicants with little or no credit history get less reliable predictions, because the model has less information about them. This matters, since these are exactly the people Home Credit wants to serve. The data also comes from a single period, so the model may not hold up as economic conditions change. With more time, we would try LightGBM, which often performs better on this kind of data, and tune more model settings. We would also build features that give recent months more weight than older ones, since recent behaviour usually says more about a person's current situation.